

Guide to Difference-in-Differences Methods: Implementation in R

TWFE, Callaway & Sant'Anna, Borusyak et al., and de Chaisemartin & D'Haultfoeuille

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Why Multiple DiD Estimators?

- **Problem:** Classic TWFE can produce biased estimates with:
 - Staggered treatment adoption
 - Heterogeneous treatment effects
 - Dynamic treatment effects
- **Solution:** Recent methodological advances provide robust alternatives
- **Goal:** Learn to implement 6 different estimators and understand when to use each

The Six Methods We'll Cover

- ① **TWFE** (Two-Way Fixed Effects) - The classic approach
- ② **Callaway & Sant'Anna (2021)** - Group-time ATTs with flexible aggregation
- ③ **Borusyak et al. (2024)** - Imputation-based estimator
- ④ **de Chaisemartin & D'Haultfoeuille (2020)** - Three variants:
 - Standard (absorbing treatment)
 - Treatment reversion (non-absorbing)
 - Intensive margin (continuous treatment)

Example of a Panel Data Structure

Your dataset: Mexican municipalities and moisture levels

Panel Dimensions:

- **Units:** 2,116 municipalities
- Why this difference vs the total municipalities in Mexico?
- **Time:** 1990-2020 (31 years)
- **Total obs:** 65,596
- **Balanced panel or Unbalanced panel?**

Treatment:

- **Type:** Staggered adoption
- **First treated:** 1991
- **Last treated:** 2020
- **Treatment share:** 31.4%

Essential Variables in Your Data

```
head(df.twfe_moisture)
#   Year INEGI_CODE firstdto tipo_municipio post_treat DTO_dummy_a
# 1 1990      1001    2002         UG           0           0
# 2 1990      1002    2004         RU           0           0
# ...
```

Variable	Type	Description
Year	Time	Year variable (1990-2020)
INEGI_CODE	Unit ID	Municipality identifier
firstdto	Treatment timing	First year of treatment (0 if never treated)
DTO_dummy_a	Treatment	Binary absorbing treatment (time-variant)

Treatment Variables: Three Types

1. Absorbing Binary Treatment (DTO_dummy_a)

- Switches from 0 to 1 once, stays at 1
- Use for: TWFE, CS, Borusyak, dCDH standard

2. Non-Absorbing Binary Treatment (Dich_Presence)

- Can switch between 0 and 1 multiple times
- Use for: dCDH treatment reversion

3. Continuous Treatment (Total_presence)

- Continuous or discrete values (intensity)
- Use for: dCDH intensive margin

Outcome Variables: Four Transformations

① IHS_Moisture_mean_weighted

- Inverse Hyperbolic Sine: $\sinh^{-1}(x) = \log(x + \sqrt{x^2 + 1})$
- Handles zeros, interpretable like logs

② Log1p_Moisture_mean_weighted

- Log transformation: $\log(1 + x)$
- Traditional approach for positive outcomes

③ Binary_Moisture_mean_weighted

- Indicator: $\mathbb{I}(x > 0)$
- For extensive margin effects

④ LogLinear_Moisture_mean_weighted

- Chen & Roth (2024) transformation for many zeros

Time-Variant vs Time-Invariant Variables

Time-Invariant (subscript i):

- $INEGI_CODE_i$ (ID)
- $tipo_municipio_i$
- $firstdto_i$
- Geographic characteristics

These are absorbed by unit fixed effects

Time-Variant (subscript it):

- $Year_t$
- $Moisture\ outcomes_{it}$
- $Treatment\ status_{it}$
- $Time_to_Treatment_{it}$

These vary over time within units

Key Point

DiD methods exploit **within-unit variation** over time while controlling for time trends

Creating Time-to-Treatment Variables

Critical for event studies: Time_to_Treatment

$$\text{Time_to_Treatment}_{it} = \text{Year}_t - \text{firstdto}_i$$

- Negative values = pre-treatment periods
- Zero = treatment year
- Positive values = post-treatment periods
- 0 for never-treated units in `firstdto`

Treatment_group: Indicator for ever-treated units (time-invariant, i)

$$\text{Treatment_group}_i = \mathbb{1}(\text{firstdto}_i > 0)$$

Data Summary Statistics - Part 1

```
summary(df.twfe_moisture)
```

	Year	INEGI_CODE	firstdto	post_treat
Min.	:1990	Min. : 1001	Min. : 0	Min. :0.00
Median	:2005	Median :20304	Median :1997	Median :0.00
Mean	:2005	Mean :19756	Mean :1279	Mean :0.31
Max.	:2020	Max. :32058	Max. :2020	Max. :1.00

Key observations:

- 64% of municipalities eventually treated
- Treatment timing varies from 1991 to 2020
- Up to 30 years of pre-treatment data

Data Summary Statistics - Part 2

Time_to_Treatment	Treatment_group	Moisture_mean_weighted
Min. : -30.00	Min. : 0.00	Min. : 0.10
Median : 0.00	Median : 1.00	Median : 0.34
Mean : -0.48	Mean : 0.64	Mean : 0.33
Max. : 29.00	Max. : 1.00	Max. : 0.45

Understanding the data structure:

- Median Time_to_Treatment = 0 means many units at treatment year
- Mean = -0.48 suggests slightly more pre-treatment observations
- Treatment_group: 64% ever-treated, 36% never-treated

TWFE: The Classic Approach

Model Specification (Roth et al. 2023, Eq. 3)

$$Y_{it} = \alpha_i + \lambda_t + \beta^{TWFE} D_{it} + \varepsilon_{it}$$

where:

- α_i = unit fixed effects (time-invariant characteristics)
- λ_t = time fixed effects (common time trends)
- D_{it} = treatment indicator
- β^{TWFE} = treatment effect (assumes homogeneity)

Key assumptions:

- 1 Parallel trends (Assumption 1, Roth et al.)
- 2 No anticipation (Assumption 2, Roth et al.)
- 3 Homogeneous treatment effects

TWFE: When Does It Work?

Works well when:

- Single treatment timing
- Homogeneous effects
- No dynamic effects
- Simple 2×2 design

Problems when:

- Staggered adoption
- Heterogeneous effects
- Dynamic treatment effects
- Uses "forbidden comparisons"

Goodman-Bacon Decomposition

TWFE with staggered timing = weighted average of good and bad comparisons. **Do you remember what "forbidden comparisons" are?**

TWFE: Forbidden Comparisons Explained

What are "forbidden comparisons"? (Goodman-Bacon 2021)

In staggered DiD, TWFE uses **already-treated units as controls**:

- Unit treated in 2014 is compared to unit treated in 2016
- In 2015, the 2014-treated unit serves as "control" for 2016-treated
- This is problematic if treatment effects are dynamic or heterogeneous

Why "forbidden"?

- Already-treated units are not valid counterfactuals
- Their outcomes reflect their own treatment effects
- Can lead to negative weights and sign reversals

TWFE: Variables Required

Variable	Purpose
INEGI_CODE	Unit identifier (for α_i)
Year	Time identifier (for λ_t)
DTO_dummy_a	Binary treatment variable
IHS_Moisture_*	Outcome variable

```
# Basic TWFE estimation
library(fixest)
twfe_model <- feols(
  IHS_Moisture_mean_weighted ~ DTO_dummy_a |
    INEGI_CODE + Year,
  data = df.twfe_moisture,
  cluster = "INEGI_CODE"
)
```

TWFE: Event Study Specification

Goal: Estimate treatment effects at each relative time period

```
# Event study TWFE (Roth et al. 2023, Eq. 16)
twfe_es <- feols(
  IHS_Moisture_mean_weighted ~
    i(Time_to_Treatment, Treatment_group, ref = -1) |
    INEGI_CODE + Year,
  data = df.twfe_moisture,
  cluster = "INEGI_CODE"
)

# Plot results
iplot(twfe_es)
```

Important:

- `ref = -1`: Sets period -1 as reference (normalized to zero)
- Tests parallel trends in pre-period
- Shows dynamic effects in post-period

TWFE: Extract Event Study Coefficients - Part 1

```
# 1. Event study estimation
plot_twfe_IHS <- feols(
  IHS_Moisture_mean_weighted ~
    i(Time_to_Treatment, Treatment_group, -1) |
    INEGI_CODE + Year,
  data = df.twfe_moisture,
  cluster = "INEGI_CODE"
)

# 2. Extract coefficients
twfe_IHS_output <- tibble(
  rowname = names(plot_twfe_IHS$coefficients),
  Estimate = plot_twfe_IHS$coefficients,
  SE = plot_twfe_IHS$se
)
```

TWFE: Extract Event Study Coefficients - Part 2

```
# Continue processing
twfe_IHS_output <- twfe_IHS_output %>%
  mutate(
    time = as.numeric(gsub(".*::", "", rowname)),
    LB_CI = Estimate - (1.96 * SE),
    UB_CI = Estimate + (1.96 * SE),
    Model = "TWFE"
  ) %>%
  filter(time >= -5 & time <= 5)

# Add reference period at -1
twfe_IHS_output <- bind_rows(
  twfe_IHS_output,
  tibble(time = -1, Estimate = 0,
          LB_CI = 0, UB_CI = 0, Model = "TWFE")
)
```

TWFE: ATT Estimation - Part 1

```
# Simple ATT (not event study)
library(lfe)
twfe_att <- felm(
  IHS_Moisture_mean_weighted ~ DTO_dummy_a |
    INEGI_CODE + Year |
    0 |
    INEGI_CODE,
  data = df.twfe_moisture
)

# Extract results
coef_twfe <- c(twfe_att$coefficients)
se_twfe <- twfe_att$cse
```

TWFE: ATT Estimation - Part 2

```
# Create results data frame
df_twfe_att <- tibble(
  type = "TWFE",
  coef = coef_twfe,
  se = se_twfe,
  conf.low = coef_twfe - 1.96 * se_twfe,
  conf.high = coef_twfe + 1.96 * se_twfe
)
```

Note: This gives a single ATT estimate assuming:

- Homogeneous treatment effects across all units
- Homogeneous treatment effects across all periods

Event Study Plot:

- Pre-trends should be flat (parallel trends test)
- Post-treatment effects show dynamics
- Reference period at -1 (zero by construction)

ATT:

- Single coefficient
- Average across all treated units/periods
- **Caution:** May be biased with heterogeneity

In your case:

- ATT = effect of DTO presence on IHS moisture
- Averaged over all municipalities
- Averaged over all post-treatment years

TWFE: Advantages and Limitations

Advantages:

- Simple and fast
- Well-understood
- Easy to implement
- Good baseline
- Works well in simple settings

Limitations:

- Biased with:
 - Staggered timing
 - Heterogeneous effects
 - Dynamic effects
- Uses "forbidden comparisons"
- Already-treated as controls

Recommendation (Roth et al. 2023)

Use TWFE as a **starting point**, but validate with robust methods

Key Innovation

Estimate treatment effects for each **(group, time)** combination separately, then aggregate

Group = Cohort that receives treatment at the same time (G_i)

Two-step procedure:

- 1 Estimate group-time ATTs: $ATT(g, t)$ for all (g, t) pairs
- 2 Aggregate across groups/times as desired

Control group options:

- Never-treated units (default)
- Not-yet-treated units (recommended when few never-treated)

CS: Why It Works Better

Problem with TWFE (Roth et al. Section 3.2)

When unit i is treated in 2005 and unit j is treated in 2010, TWFE uses j as a control for i in 2007-2009, even though j is being "pulled toward" treatment

CS Solution (Assumption 4, Roth et al.)

Only compare units in group g (treated at time g) to:

- Never-treated units, OR
- Units not yet treated at time t

This avoids using already-treated units as controls = **clean comparisons**

CS: Variables Required

Variable	Purpose
<code>idname</code>	Unit identifier (INEGI_CODE)
<code>tname</code>	Time variable (Year)
<code>gname</code>	First treatment period (<code>firstdto</code>)
<code>yname</code>	Outcome variable
<code>xformula</code>	Covariates (optional)

Key requirement for `gname`:

- First period of treatment for treated units
- 0 for never-treated units (in your `firstdto` variable)

CS: Basic Implementation - Part 1

```
library(did)

# Step 1: Estimate group-time ATTs
cs_IHS <- att_gt(
  yname = "IHS_Moisture_mean_weighted",
  gname = "firstdto",      # Treatment cohort
  idname = "INEGI_CODE",   # Unit ID
  tname = "Year",          # Time variable
  xformla = ~1,            # No covariates
  control_group = "notyettreated",
  data = df.twfe_moisture,
  est_method = "reg",
  base_period = "universal"
)
```

Understanding the arguments:

- `control_group = "notyettreated"`: Uses not-yet-treated as controls (better when few never-treated)
- `base_period = "universal"`: Uses all pre-periods for comparison, but normalized to -1
- `est_method = "reg"`: Regression-based (alternatives: "ipw", "dr")

Output: Estimates $ATT(g, t)$ for each cohort g at each time t

CS: Aggregation Options

```
# 1. Dynamic aggregation (event study)
# Roth et al. Eq. (9)
cs_IHS_dynamic <- aggte(
  cs_IHS,
  type = "dynamic",
  bstrap = FALSE,
  cband = FALSE,
  na.rm = TRUE
)

# 2. Simple aggregation (overall ATT)
cs_IHS_simple <- aggte(
  cs_IHS,
  type = "simple",
  na.rm = TRUE
)

# 3. Group-specific aggregation
cs_IHS_group <- aggte(cs_IHS, type = "group")
```

CS: Extract Event Study Results - Part 1

```
# Extract dynamic effects for plotting
cs_IHS_output <- tibble(
  time = cs_IHS_dynamic$egt,
  Estimate = cs_IHS_dynamic$att.egt,
  SE = cs_IHS_dynamic$se.egt
)
```

CS: Extract Event Study Results - Part 2

```
# Continue processing
cs_IHS_output <- cs_IHS_output %>%
  filter(time >= -5 & time <= 5) %>%
  mutate(
    LB_CI = Estimate - (1.96 * SE),
    UB_CI = Estimate + (1.96 * SE),
    Model = "CS"
  ) %>%
  select(-SE)

# Built-in plotting
ggdid(cs_IHS_dynamic)
```

CS: Extract Overall ATT

```
# Simple aggregation for overall ATT
agg_att_cs_IHS <- aggte(
  cs_IHS,
  type = "simple",
  na.rm = TRUE
)

# Extract results
df_cs_att_IHS <- tibble(
  type = "Callaway & Sant'Anna",
  coef = agg_att_cs_IHS$overall.att,
  se = agg_att_cs_IHS$overall.se,
  conf.low = agg_att_cs_IHS$overall.att -
    1.96 * agg_att_cs_IHS$overall.se,
  conf.high = agg_att_cs_IHS$overall.att +
    1.96 * agg_att_cs_IHS$overall.se
)
```

The `est_method` argument offers different estimators:

Method	Description
"reg"	Outcome regression (default, Section 4.2)
"ipw"	Inverse probability weighting (Section 4.2)
"dr"	Doubly robust (recommended, Section 4.2)

Doubly robust (Sant'Anna & Zhao 2020):

- Consistent if either outcome model OR propensity score model is correct
- More robust to model misspecification
- Achieves semiparametric efficiency bound when both models correct

CS: Interpreting Results

Event Study Interpretation:

- Each point = average effect at ℓ periods relative to treatment
- Aggregated across all cohorts using specified weights
- Pre-treatment estimates test parallel trends
- Post-treatment estimates show dynamic effects

Overall ATT Interpretation:

- Weighted average across all (g, t) combinations
- Weights based on group size and exposure time
- More reliable than TWFE with heterogeneity

In your application:

- Effect of DTO presence on moisture, allowing effects to differ by cohort and time

CS: Advantages and Limitations

Advantages:

- Robust to heterogeneous effects
- Handles staggered adoption
- Flexible aggregation
- Doubly-robust option
- Well-documented package

Limitations:

- Requires sufficient never/not-yet-treated units
- Can be slow with large datasets
- Requires absorbing treatment

When to Use (Roth et al. Section 3.5)

Your data has staggered adoption and you want robust estimates with clean comparisons

Key Innovation (Roth et al. Section 3.3)

Use **never-treated** or **not-yet-treated** units to impute counterfactual outcomes for treated units

Intuition:

- 1 Estimate unit and time fixed effects using **only** untreated observations
- 2 Predict (impute) what would have happened to treated units without treatment
- 3 Compare actual outcomes to imputed counterfactuals

Result: Unbiased estimates even with heterogeneous and dynamic effects

Imputation Formula

$$\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$$

where $\hat{Y}_{it}(0)$ is imputed from:

$$\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$$

and $\hat{\alpha}_i, \hat{\lambda}_t$ are estimated using only untreated observations

Key difference from TWFE: Fixed effects estimated only on control units, not on treated units post-treatment

Borusyak: Variables Required

Variable	Purpose
data	Your dataset
yname	Outcome variable (string)
gname	First treatment period
tname	Time variable (string)
idname	Unit identifier (string)
cluster_var	Clustering variable

Critical for gname: Must equal 0 for never-treated units (as in your firstdto)

Borusyak: Implementation Strategy

Your pipeline uses a split approach:

- 1 Estimate **post-treatment** effects separately
- 2 Estimate **pre-treatment** placebos separately
- 3 Combine for event study

Why? Better control over time horizons and cleaner results

```
library(didimputation)

# Post-treatment effects
borus_post <- did_imputation(
  data = df.twfe_moisture,
  yname = "IHS_Moisture_mean_weighted",
  gname = "firstdto",
  first_stage = ~ 0 | Year + INEGI_CODE,
  tname = "Year",
  idname = "INEGI_CODE",
  cluster_var = "INEGI_CODE",
  horizon = c(0:5)
```

Borusyak: Pre-treatment Placebos

```
# Pre-treatment (placebo) effects
borus_pre <- did_imputation(
  data = df.twfe_moisture,
  yname = "IHS_Moisture_mean_weighted",
  gname = "firstdto",
  first_stage = ~ 0 | Year + INEGI_CODE,
  tname = "Year",
  idname = "INEGI_CODE",
  cluster_var = "INEGI_CODE",
  pretrends = c(-5:-1)
)
```

Why test pre-trends?

- Validates parallel trends assumption (Roth et al. Section 4.3)
- Should be close to zero
- Significant pre-trends = violation of identification

Borusyak: Format Post-Treatment Results

```
# Format post-treatment results
borus_post_output <- borus_post %>%
  mutate(
    time = as.numeric(term),
    Model = "Borusyak Post",
    Estimate = estimate,
    LB_CI = conf.low,
    UB_CI = conf.high
  ) %>%
  select(time, Estimate, LB_CI, UB_CI, Model) %>%
  filter(time <= 5)
```

Borusyak: Format Pre-Treatment Results

```
# Format pre-treatment results
borus_pre_output <- borus_pre %>%
  mutate(
    time = as.numeric(term),
    Model = "Borusyak Pre",
    Estimate = estimate,
    LB_CI = conf.low,
    UB_CI = conf.high
  ) %>%
  select(time, Estimate, LB_CI, UB_CI, Model) %>%
  filter(time >= -5)
```

Next: Combine these for visualization and hypothesis testing

Borusyak: Overall ATT

```
# Overall ATT (no horizon specified)
borus_att <- did_imputation(
  data = df.twfe_moisture,
  yname = "IHS_Moisture_mean_weighted",
  gname = "firstdto",
  first_stage = ~ 0 | Year + INEGI_CODE,
  tname = "Year",
  idname = "INEGI_CODE",
  cluster_var = "INEGI_CODE"
) %>%
mutate(type = "Borusyak et al.") %>%
select(type, coef = estimate, se = std.error,
       conf.low, conf.high)
```

Borusyak: Creating Event Study Plot - Part 1

```
# Set consistent y-axis limits
y_min <- min(c(
  min(borus_pre_output$LB_CI),
  min(borus_post_output$LB_CI)
))
y_max <- max(c(
  max(borus_pre_output$UB_CI),
  max(borus_post_output$UB_CI)
))
```

Borusyak: Creating Event Study Plot - Part 2

```
# Placebo plot (pre-treatment)
placebo_plot <- ggplot(
  borus_pre_output,
  aes(x = time, y = Estimate)
) +
  geom_point(size = 3, color = "black") +
  geom_errorbar(
    aes(ymin = LB_CI, ymax = UB_CI),
    width = 0.2, color = "black"
  ) +
  geom_hline(yintercept = 0,
             linetype = "dashed") +
  geom_vline(xintercept = -1,
             linetype = "dotted") +
  scale_y_continuous(limits = c(y_min, y_max))
```

Borusyak: Creating Event Study Plot - Part 3

```
# Treatment effects plot
treatment_plot <- ggplot(
  borus_post_output,
  aes(x = time, y = Estimate)
) +
  geom_point(size = 3, color = "black") +
  geom_errorbar(
    aes(ymin = LB_CI, ymax = UB_CI),
    width = 0.2, color = "black"
  ) +
  geom_hline(yintercept = 0,
             linetype = "dashed") +
  geom_vline(xintercept = 0,
             linetype = "dotted") +
  scale_y_continuous(
    limits = c(y_min, y_max),
    position = "right"
  )
```

Borusyak: Combine Plots

```
# Combine using patchwork  
library(patchwork)  
plot_borus <- placebo_plot + treatment_plot
```

Result: Side-by-side visualization of:

- Left panel: Pre-treatment placebos (should be near zero)
- Right panel: Post-treatment effects (actual treatment effects)
- Same y-axis scale for direct comparison

Pre-treatment plot:

- Should hover around zero
- Tests parallel trends assumption (Assumption 4, Roth et al.)
- Significant deviations suggest violation

Post-treatment plot:

- Shows effect at each period after treatment
- Allows for time-varying treatment effects
- Each estimate is treatment effect at that specific lag

Overall ATT:

- Weighted average of all post-treatment effects
- Weights proportional to number of treated units at each horizon

Borusyak vs CS: Key Differences

Main difference: How pre-treatment periods are used

Borusyak:

- Uses ALL pre-periods
- Averages over all pre-treatment data
- More efficient if errors not serially correlated
- Requires parallel trends for all periods (Assumption 4)

CS:

- Uses last pre-period only
- Default: period $g - 1$ as baseline
- More robust to long-run trends
- Weaker parallel trends assumption (Assumption 4.a)

Trade-off: Efficiency vs. robustness to specification

Borusyak: Advantages and Limitations

Advantages:

- Computationally fast
- Handles heterogeneity
- Clean imputation logic
- Good pre-trend testing
- Flexible horizons
- Efficient under homoskedasticity

Limitations:

- Requires untreated units
- Assumes linear FE model
- Less flexible than CS
- Limited covariate control
- Stronger parallel trends assumption

When to Use (Roth et al. Section 3.3)

You want a fast, intuitive estimator and are confident in parallel trends across all periods

dCDH: Overview of Three Variants

de Chaisemartin & D'Haultfoeuille (2020, 2024) provide three estimators:

① **Standard (Absorbing):** Binary treatment that persists

- Uses `DT0_dummy_a`
- Like CS and Borusyak, but different methodology

② **Treatment Reversion:** Treatment can turn on/off

- Uses `Dich_Presence`
- Allows treatment reversals (Section 3.4, Roth et al.)

③ **Intensive Margin:** Continuous treatment dose

- Uses `Total_presence`
- Estimates effect per unit of treatment

dCDH Standard: Key Features

Main Innovation

Explicitly accounts for **dynamic treatment effects** and **heterogeneity** across groups and time

What makes it different?

- Uses "switchers" vs "non-switchers" comparisons
- Explicitly models first-difference-in-differences
- Provides built-in placebo tests
- Fast analytical standard errors (no bootstrap needed)

Compatible with:

- Staggered adoption
- Heterogeneous effects over time
- Dynamic effects

dCDH Standard: Variables Required

Variable	Purpose
df	Dataset
outcome	Outcome variable (must be numeric)
group	Unit identifier (must be numeric)
time	Time variable (must be numeric)
treatment	Binary absorbing treatment (numeric)
cluster	Clustering variable (optional)

```
library(DIDmultiplegtDYN)

dcdh_standard <- did_multiplegt_dyn(
  df = df.twfe_moisture,
  outcome = "IHS_Moisture_mean_weighted",
  group = "INEGI_CODE",
  time = "Year",
  treatment = "DTO_dummy_a",
  effects = 50,
  placebo = 50,
```

dCDH Standard: Full Implementation

```
# Estimate with clustering
dcdh_IHS <- did_multiplegt_dyn(
  df = df.twfe_moisture,
  outcome = "IHS_Moisture_mean_weighted",
  group = "INEGI_CODE",
  time = "Year",
  treatment = "DTO_dummy_a",
  placebo = 50,          # Request 50 pre-periods
  effects = 50,          # Request 50 post-periods
  graph_off = TRUE,     # Suppress automatic graph
  cluster = "INEGI_CODE" # Cluster SEs
)

# Note: Returns as many periods as available
# in data (not necessarily 50)
```

dCDH Standard: Extract Event Study Results

```
# Extract event study results
dcdh_IHS_output <- as.data.frame(
  dcdh_IHS[["plot"]][["data"]]
) %>%
  rename(
    time = "Time",
    Estimate = "Estimate",
    LB_CI = "LB.CI",
    UB_CI = "UB.CI"
  ) %>%
  mutate(
    Model = "dCDH",
    time = time - 1 # CRITICAL: Adjust time index
  ) %>%
  filter(time >= -5 & time <= 5)
```

dCDH Standard: Extract Overall ATE

```
# Extract overall ATE
coef_dcdh <- as.data.frame(dcdh_IHS$results$ATE)

df_dcdh_att <- coef_dcdh %>%
  select(Estimate, SE, 'LB CI', 'UB CI') %>%
  mutate(
    type = "de Chaisemartin & D'Haultfoeuille"
  ) %>%
  rename(
    coef = "Estimate",
    se = "SE",
    conf.low = "LB CI",
    conf.high = "UB CI"
  )
```

dCDH Standard: Understanding Output

The function returns a list with several components:

Component	Contains
results	Scalar results (ATE, placebos)
plot	ggplot object with event study
coef	All coefficient estimates
args	Arguments used in estimation

Automatic output includes:

- Event study estimates (effects)
- Pre-treatment placebos
- Average total effect
- Joint test of placebo nullity (p-value)

dCDH Standard: Accessing Built-in Plot

```
# The package creates a ggplot object
plot_dcdh <- dcdh_IHS$plot

# Customize using ggplot2
library(ggplot2)
plot_dcdh +
  labs(title = "dCDH Event Study") +
  theme_minimal() +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "red")

# Or extract data and create custom plot
plot_data <- dcdh_IHS$plot$data
```

dCDH Standard: Interpreting Results

Event study effects:

- Effect at ℓ periods after first treatment
- Averaged across all switchers at that horizon
- Allows for heterogeneity across cohorts

Placebos:

- Pre-treatment "effects" (should be zero)
- Joint test p-value tests parallel trends
- Significant placebos = parallel trends violation

Average total effect:

- Weighted average of all post-treatment effects
- Weights based on number of switchers and periods

dCDH Standard: Time Index Adjustment

Important Detail

dCDH uses different time indexing: **always** adjust by subtracting 1

Period	dCDH Output	Standard (Your Code)
2 before treatment	Time = 1	time = 0 (after adjustment)
1 before treatment	Time = 2	time = 1 (after adjustment)
Treatment period	Time = 3	time = 2 (after adjustment)
1 after treatment	Time = 4	time = 3 (after adjustment)
2 after treatment	Time = 5	time = 4 (after adjustment)

In your code: `mutate(time = time - 1)`

dCDH Reversion: When Treatment Reverses

Motivation (Roth et al. Section 3.4)

Sometimes treatment is **not absorbing**:

- Policies can be repealed
- Programs can be discontinued
- Interventions can be temporary

Example from your data: Dich_Presence

- $D_{it} = 1$ when DTO is present in municipality i at time t
- Can switch back to 0 if DTO leaves
- Can switch multiple times

Why this matters: Standard methods (TWFE, CS, Borusyak) assume treatment persists!

dCDH Reversion: Key Differences

Standard (Absorbing):

- $D_{it} \in \{0, 1\}$
- Once treated, always treated
- $D_{it} = 0 \Rightarrow D_{it+1} \geq D_{it}$
- Uses first treatment timing

Reversion (Non-absorbing):

- $D_{it} \in \{0, 1\}$
- Treatment can reverse
- D_{it+1} can be $< D_{it}$
- Uses current treatment status

Critical Point

Must use a **time-varying** treatment variable, not first treatment date

dCDH Reversion: Implementation

```
# Use non-absorbing treatment variable
dcdh_reversion_IHS <- did_multiplegt_dyn(
  df = df.twfe_moisture,
  outcome = "IHS_Moisture_mean_weighted",
  group = "INEGI_CODE",
  time = "Year",
  treatment = "Dich_Presence", # Non-absorbing!
  effects = 50,
  placebo = 50,
  graph_off = TRUE,
  cluster = "INEGI_CODE"
)
```

Key change: treatment = "Dich_Presence" instead of DTO_dummy_a

dCDH Reversion: Extract Results

```
# Extract event study results
dcdh_reversion_IHS_output <- as.data.frame(
  dcdh_reversion_IHS[["plot"]][["data"]]
) %>%
  rename(
    time = "Time",
    Estimate = "Estimate",
    LB_CI = "LB.CI",
    UB_CI = "UB.CI"
  ) %>%
  mutate(
    Model = "dCDH Reversion",
    time = time - 1 # Adjust time index
  ) %>%
  filter(time >= -5 & time <= 5)

# Extract ATE
coef_dcdh_reversion <- as.data.frame(
  dcdh_reversion_IHS$results$ATE
```

dCDH Reversion: Interpreting Results

Treatment effect now captures:

- Effect of *being treated* at time t
- **Not** effect of *first treatment*
- Includes units that may have been treated before

Comparison groups:

- Units not treated at time t
- May include units that were treated before but reverted
- May include units that will be treated later

Interpretation example:

"Being under treatment at time t increases outcome by β relative to not being under treatment at time t "

dCDH Reversion: Additional Assumptions

No Carryover Assumption (dCDH 2020, Roth et al. Section 3.4):

Potential outcomes depend only on **current** treatment status, not treatment history:

$$Y_{it}(d_1, d_2, \dots, d_t) = Y_{it}(d_t)$$

When this may be violated:

- Treatment has lasting effects after removal
- Example: Education policy removed, but knowledge remains
- Example: Infrastructure built, doesn't disappear when policy ends

Extensions: dCDH (2022a) relax this with stronger parallel trends

dCDH Reversion: When to Use

Use reversion estimator when:

- Treatment is genuinely non-absorbing
- Interest in contemporaneous effect of treatment
- Treatment can be discontinued/reversed
- Policy evaluation where policy can be repealed
- Willing to assume no carryover effects

Examples:

- Temporary policy interventions
- Presence of organization/program that can leave
- Membership in trade agreements
- Employment in certain sectors
- **Your case:** DTO presence/absence in municipality

Motivation

Often treatment is not binary but varies in **intensity**:

- Dose of medication
- Amount of aid/funding
- Number of programs/facilities
- **Your case**: Number of DTOs present

Goal: Estimate effect *per unit* of treatment

$$\text{Effect} = \beta \times (\text{Treatment dose})$$

Your variable: `Total_presence` = count of DTOs in municipality

Extensive Margin:

- Any treatment vs none
- Binary: $D \in \{0, 1\}$
- "Treated vs untreated"

Effect of having treatment

Intensive Margin:

- How much treatment
- Continuous: $D \in [0, \infty)$
- "Dose-response"

Effect per unit of treatment

Interpretation Difference

Extensive: "Having a DTO increases moisture by β "

Intensive: "Each additional DTO increases moisture by β "

dCDH Intensive: Implementation

```
# Use continuous treatment variable
dcdh_intensive_IHS <- did_multiplegt_dyn(
  df = df.twfe_moisture,
  outcome = "IHS_Moisture_mean_weighted",
  group = "INEGI_CODE",
  time = "Year",
  treatment = "Total_presence", # Continuous!
  effects = 50,
  placebo = 50,
  graph_off = TRUE,
  cluster = "INEGI_CODE"
)
```

Key change: treatment = "Total_presence" (continuous/count variable)

dCDH Intensive: Extract Results

```
# Extract event study results
dcdh_intensive_IHS_output <- as.data.frame(
  dcdh_intensive_IHS[["plot"]][["data"]]
) %>%
  rename(
    time = "Time",
    Estimate = "Estimate",
    LB_CI = "LB.CI",
    UB_CI = "UB.CI"
  ) %>%
  mutate(
    Model = "dCDH Intensive",
    time = time - 1 # Adjust time index
  ) %>%
  filter(time >= -5 & time <= 5)

# Extract ATE per unit
coef_dcdh_intensive <- as.data.frame(
  dcdh_intensive_IHS$results$ATE
```

Standard coefficient interpretation:

"A one-unit increase in treatment increases the outcome by β "

In your context:

- If $\hat{\beta} = 0.05$ for IHS moisture
- Interpretation: "Each additional DTO present increases IHS moisture by 0.05 units"
- Dynamic effects show how this effect evolves over time since treatment change

Scaling:

- Can multiply by actual treatment dose changes
- E.g., going from 1 to 3 DTOs: $\Delta D = 2$, so effect = $2 \times \hat{\beta}$

Strong Linearity Assumption

The intensive margin estimator assumes **linear dose-response**:

- Effect of $0 \rightarrow 1$ = effect of $1 \rightarrow 2$ = effect of $2 \rightarrow 3$
- No saturation or threshold effects
- No diminishing/increasing returns

When this may fail:

- Diminishing returns to treatment (e.g., 10th DTO has less effect than 1st)
- Threshold effects (need minimum dose for any effect)
- Non-linear dose-response relationship

Robustness check: Compare to binary extensive margin estimates. If very different, linearity may be violated.

When to use intensive margin:

- Treatment naturally varies in intensity
- Interest in dose-response relationship
- Policy question about "how much" treatment
- Want to optimize treatment levels
- Comfortable assuming linear dose-response

Advantages:

- More statistical power (uses variation in doses, not just on/off)
- Policy-relevant estimates (cost-benefit analysis)
- Can predict effects of scaling up/down
- Answers "how much" rather than just "whether"

Summary: Which Method When?

Method	Treatment Type	Best For	Key Advantage
TWFE	Binary/Absorbing	Baseline	Fast, simple
Callaway-Sant'Anna	Binary/Absorbing	Staggered timing	Clean comparison groups
Borusyak	Binary/Absorbing	Fast computation	Imputation intuition
dCDH Standard	Binary/Absorbing	Robust alternative	Fast, no bootstrap
dCDH Reversion	Binary/Non-absorbing	Treatment reversal	Allows switching off
dCDH Intensive	Continuous	Dose-response	Effect per unit

Recommended Workflow (Roth et al. Section 3.5)

1 Start with TWFE

- Quick baseline
- Check event study for obvious issues

2 Implement all robust methods

- CS, Borusyak, AND dCDH standard
- Check if results are consistent

3 Consider special features

- If treatment reverses: dCDH reversion
- If continuous treatment: dCDH intensive

4 Report all estimates

- Show robustness
- Discuss any differences

Creating Comparison Plots

Your pipeline creates three types of comparisons:

1. Event studies for each method

- Separate plot per method
- Shows pre-trends and dynamic effects
- Easy to compare parallel trends tests

2. Combined ATT plot

- All methods' overall ATTs on one plot
- Shows point estimates and confidence intervals
- Highlights agreement/disagreement across methods

3. Faceted by transformation

- Separate panel for each outcome transformation
- Shows robustness to outcome specification

Combining Results - Part 1

```
# Add transformation identifier
att_results_IHS$transformation <- "IHS"
att_results_Log1p$transformation <- "Log(1+x)"
att_results_Binary$transformation <- "Binary"
att_results_CR$transformation <- "Chen-Roth"

# Combine all results
att_results_all <- bind_rows(
  att_results_IHS,
  att_results_Log1p,
  att_results_Binary,
  att_results_CR
)
```

Combining Results - Part 2

```
# Create faceted plot
ggplot(att_results_all,
       aes(x = coef, y = type)) +
  geom_point(size = 3) +
  geom_errorbarh(
    aes(xmin = conf.low, xmax = conf.high),
    height = 0.2
  ) +
  geom_vline(xintercept = 0,
             linetype = "dashed") +
  facet_wrap(~ transformation,
            scales = "free_x",
            ncol = 2)
```

What If Methods Disagree? (Roth et al. Section 3.5)

If robust methods give different estimates:

① Check pre-trends

- Which method has flattest pre-trends?
- Are parallel trends violated?

② Examine treatment heterogeneity

- Look at group-specific or time-specific effects
- Some methods may weight groups differently

③ Consider sample composition

- Do you have enough never-treated units?
- Is treatment highly staggered?

④ Report range of estimates

- Show sensitivity to method choice
- Discuss which estimate is most credible and why

Common Pitfalls and Solutions

Pitfall	Solution
Few never-treated units	Use "notyettreated" control group in CS
Very staggered timing	Limit event study window (-5 to +5)
Treatment reversal	Use dCDH reversion estimator
Continuous treatment	Use dCDH intensive margin
Pre-trends violation	Re-examine identification, add co-variates, sensitivity analysis (Section 4)
Different results across methods	Report all, discuss which assumptions most credible

Outcome Transformation Considerations

Your pipeline uses 4 transformations:

- ① **IHS:** Good general-purpose transformation
 - Handles zeros, interpretable like logs
- ② **Log(1+x):** Traditional approach
 - Percentage change interpretation
- ③ **Binary:** Extensive margin
 - Effect on probability of any outcome
- ④ **Chen-Roth:** For many zeros
 - Robust to excess zeros

Best practice: Results should be qualitatively similar across transformations. If not, parallel trends may be sensitive to functional form (Roth & Sant'Anna 2023).

Standard Error Clustering (Roth et al. Section 5.2)

Always cluster at unit level:

```
# TWFE
feols(..., cluster = "INEGI_CODE")

# CS
att_gt(..., clustervars = "INEGI_CODE")

# Borusyak
did_imputation(..., cluster_var = "INEGI_CODE")

# dCDH
did_multiplegt_dyn(..., cluster = "INEGI_CODE")
```

Why?

- Accounts for serial correlation within units
- More conservative standard errors
- Standard practice in panel DiD

Pre-Trend Testing (Roth et al. Section 4.3-4.4)

All methods should show flat pre-trends:

- Visual inspection: event study plot
- Statistical test: joint test of pre-treatment coefficients
- dCDH provides automatic joint test

What if pre-trends are violated?

- ① Add covariates (if available) - Section 4.2
- ② Use shorter pre-period window
- ③ Consider group-specific trends
- ④ Sensitivity analysis (Rambachan & Roth 2022) - Section 4.5
- ⑤ May need different identification strategy

Critical

If pre-trends fail, DiD identification is questionable. Consider sensitivity analyses.

Essential components:

① Event study plots

- Show pre-trends and dynamic effects
- Include confidence intervals (preferably simultaneous)
- Mark reference period clearly

② ATT table

- All methods in one table
- Point estimates, SEs, CIs
- Sample sizes and number of treated units

③ Robustness checks

- Different outcome transformations
- Different time windows
- Sensitivity to method choice
- Pre-trend power analysis (Roth 2022)

Saving Your Results - Part 1

```
# Save event study plots
ggsave("IHS_twfe_event_study.png",
       plot_twfe_IHS,
       width = 8, height = 6, dpi = 300)

ggsave("IHS_cs_event_study.png",
       plot_cs_IHS,
       width = 8, height = 6, dpi = 300)

ggsave("IHS_borus_event_study.png",
       plot_borus_IHS,
       width = 8, height = 6, dpi = 300)
```

Saving Your Results - Part 2

```
# Save ATT results
write.csv(att_results_all,
          "att_results_all_methods.csv",
          row.names = FALSE)

# Save combined plot
ggsave("ATT_all_transformations_methods.png",
        att_plot_all,
        width = 14, height = 10, dpi = 300)
```

Key Takeaways

- ① **TWFE is biased** with staggered timing and heterogeneous effects (Goodman-Bacon 2021)
- ② **Multiple robust methods** now available:
 - CS: Clean comparisons with flexible aggregation
 - Borusyak: Fast imputation-based estimator
 - dCDH: Versatile, handles various treatment types
- ③ **Choose method** based on treatment type:
 - Absorbing binary → CS, Borusyak, or dCDH standard
 - Reversible → dCDH reversion
 - Continuous → dCDH intensive
- ④ **Report multiple estimates** for robustness
- ⑤ **Test pre-trends and conduct sensitivity analyses**

Your Complete Pipeline

What your code does:

- Implements 6 DiD estimators
- Tests 4 outcome transformations
- Creates individual event study plots
- Generates combined ATT comparison plots
- Exports publication-ready figures

Total outputs:

- 24 event study plots (6 methods $\times$ 4 transformations)
- 4 ATT comparison plots (1 per transformation)
- 1 grand comparison (all methods $\times$ all transformations)

Resources for Further Learning

Documentation:

- CS: <https://bcallaway11.github.io/did/>
- Borusyak: https://github.com/borusyak/did_imputation
- dCDH: <https://github.com/chaisemartinPackages/DIDmultiplegtDYN>

Key papers:

- **Roth et al. (2023, JoE)** - Comprehensive synthesis
- Callaway & Sant'Anna (2021, JoE)
- Borusyak, Jaravel, & Spiess (2024, REStud)
- de Chaisemartin & D'Haultfoeuille (2020, AER)
- Goodman-Bacon (2021, JoE)

Online resources:

- <https://asjadnaqvi.github.io/DiD/>

Best Practices

- ① Always check pre-trends visually and statistically
- ② Use multiple robust estimators (not just TWFE)
- ③ Cluster standard errors at unit level
- ④ Report full event studies, not just ATTs
- ⑤ Show robustness to method and specification choices
- ⑥ Be transparent about any pre-trend violations
- ⑦ Conduct sensitivity analyses when parallel trends uncertain

Questions?